Student Academic Performance Prediction using Linear Regression

Implementation Code

```
import pandas as pd
from sklearn.linear_model import LinearRegression

# Sample training data: study hours and marks
data = {
    "StudyHours": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    "Marks": [20, 25, 35, 40, 50, 55, 65, 70, 80, 90]
}

df = pd.DataFrame(data)
X = df[["StudyHours"]]
y = df["Marks"]

# Train the Linear Regression model
model = LinearRegression()
model.fit(X, y)

print("STUDENT PERFORMANCE PREDICTION")
print("-------------------------------")

try:
    hours = float(input("Enter study hours: "))

    if hours < 0:
        print("Study hours cannot be negative.")
    else:
        predicted_marks = model.predict(
            pd.DataFrame({"StudyHours": [hours]})
        )[0]
        predicted_marks = max(0, min(100, predicted_marks))

        print("Study Hours:", hours)
        print("Predicted Marks:", round(predicted_marks, 2))
        print("Note: This is an estimate from sample data.")
except ValueError:
    print("Please enter a valid number.")
```

Output (Step by step Screenshot):

Fig. 1 shows the sample dataset used to train the model.

Sample Dataset (pandas DataFrame)

Index	StudyHours	Marks
0	1	20
1	2	25
2	3	35
3	4	40
4	5	50
5	6	55
6	7	65
7	8	70
8	9	80
9	10	90

Fig. 1: Program output for 6 study hours

Fig. 2 shows an example of the program output when the user enters 6 study hours.

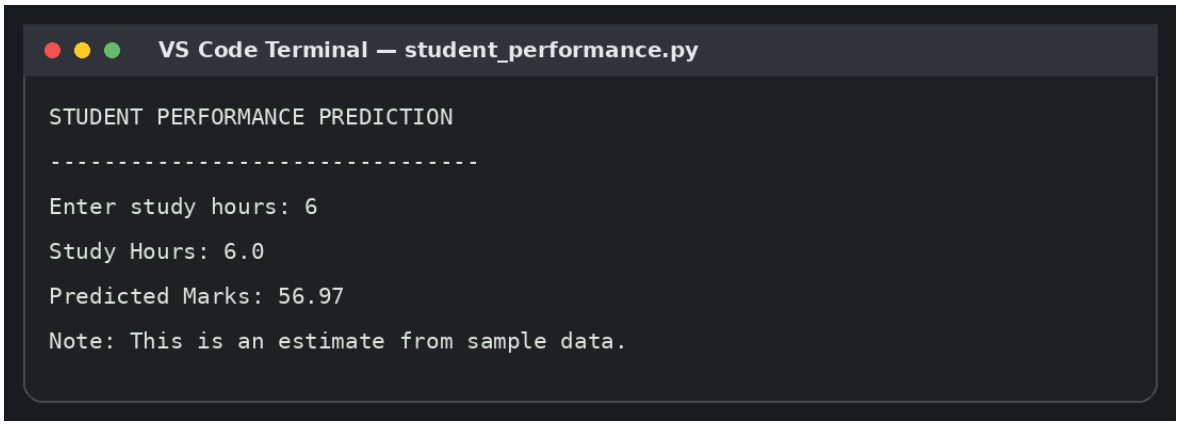


Fig. 2: Program output for 6 study hours

Fig. 3 presents the sample data and the fitted Linear Regression line. The graph is illustrative and is generated from the sample values.

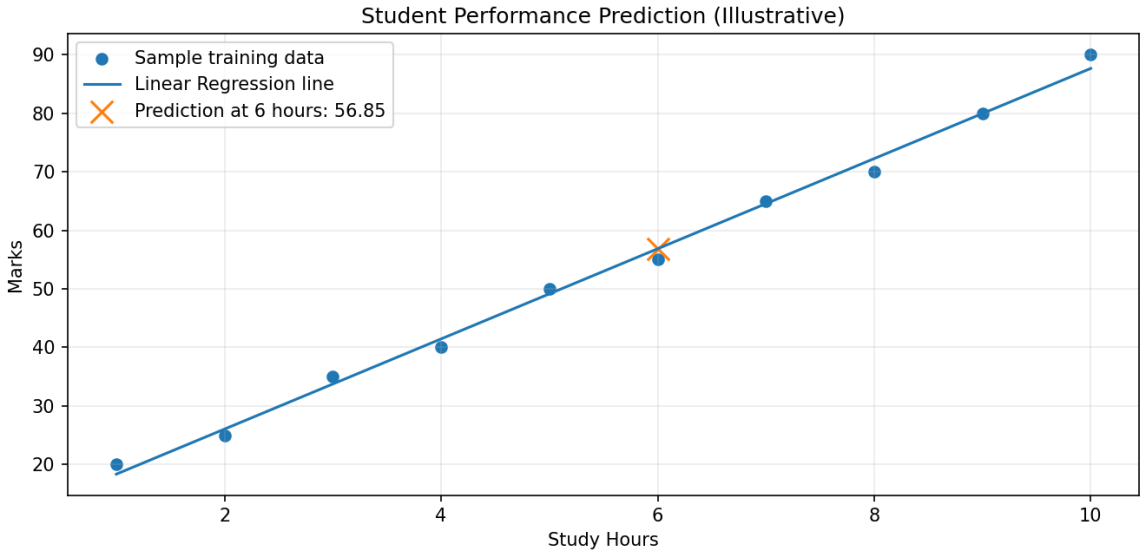


Fig. 3: Linear Regression prediction graph

Fig. 4 shows how the program handles negative study hours or invalid text.

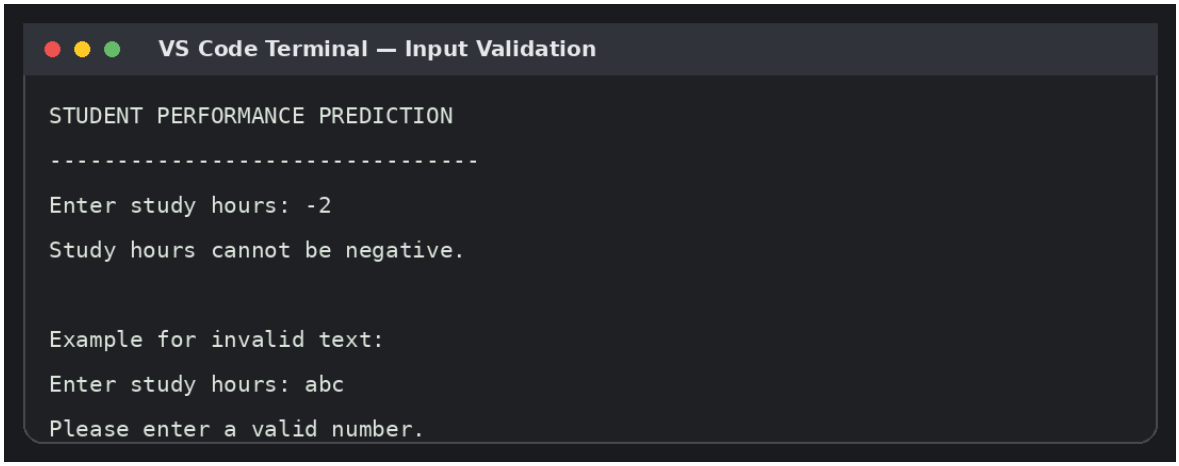


Fig. 4: Input validation output